

CHE 205, Fall 2019, Problem Set 4

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Due date: 6:00pm, Sun, 10/06/2019

All problems are from the textbook by V.L. Law.

Submit three files (spreadsheets), one for each problem, on blackboard.

Problem 1 (8 pts). Exercise 3.1.

Problem 2 (8 pts). Exercise 3.3.

Problem 3 (8 pts). Exercise 3.4.

Exercise 3.1: Mass balance on a gas absorber

A gas absorber is fed, via stream F_1 , 100 mol/min of monoethanolamine (MEA) and CO_2 . Stream F_1 is composed of 98% (mol%) MEA and 2% CO_2 . Stream F_2 contains CO_2 , SO_2 , and N_2 (Figure 3.1). Experimental data available for the unit are shown in Table 3.2.

Derive the linear system for the absorber using the supplied information. The three unknowns are the flow rates of P_1 , F_2 , and P_2 . Note that since data are available on only two components, it is necessary to include the overall material balance as one of the equations.

- Using SYSLIN (or the linear system option from Matrix.xla)
- By calculating A^{-1} followed by multiplication ($A^{-1}b$)
- By calculating the pseudo-inverse, A^+ , followed by matrix multiplication A^+b

Exercise 3.3: Flash tanks in series

Shown in Figure 3.3 is a schematic of a separation system consisting of two flash tanks in series. Experimental data are available on streams F , V_1 , V_2 , and L_2 as shown in Table 3.3. The feed rate, F , is 1000 kg/min.

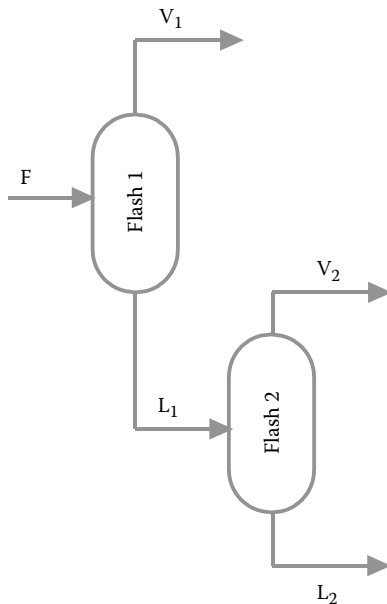


FIGURE 3.3 Two flash tanks in series.

TABLE 3.3
Stream Mass Fractions

Component	Mass Fraction			
	F	V ₁	V ₂	L
Methanol	0.3	0.71	0.44	0.08
Ethanol	0.4	0.27	0.55	0.39
Butanol	0.3	0.02	0.01	0.53

Write mass balances on each of the three components using the supplied data. The result is three linear algebraic equations in the three unknown flowrates, V_1 , V_2 , and L_2 . Use any method to find these three unknowns.

Exercise 3.4: For the following exercises, use the random matrix generator available in Matrix.xla as required.

- Generate a random matrix of size 3×3 and a column vector of length 3. Using these data to represent A and b in the linear system $Ax = b$, solve the system
 - By taking the matrix inverse followed by matrix-vector multiplication
 - Use the SYSLIN function
 - Use the function MPseudoinv and matrix multiplication
- Repeat part a for a 10×10 matrix and associated vector.
- Repeat step 3 of part a for a 4×3 matrix.
- Repeat step 3 of part a for a 3×4 matrix.
- For the 3×4 matrix of part d, apply Equation 3.53 to generate a particular solution based on an arbitrary vector, $z = [1 \ 1 \ 1 \ 1]^T$.

Note that for a 3×4 matrix, A^+A is not equal to the identity matrix. An infinite number of solutions exist since *any* z gives a proper solution.